Safety - Electrical

If you are inexperienced with the use of equipment such as oscilloscope, multimeter, function generator and power supply.

Please consult the laboratory technician to ensure safe operation.

Safety Precautions -Electrical

#1) Do not use any faulty or malfunctioning equipment.
Report it to the technician immediately.

#2) Don’t use devices with frayed cords, broken plugs, or damaged insulation.

#3) Do not overload the sockets or extension cord.

#4) If you sense burning smell, report it to the technician immediately.

#5) Keep drinking water away from the operating equipment.

#6) Switch off the socket when the equipment is not in used.

#7) Use only equipment with the Singapore SAFETY Mark.

SAFETY AND HEALTH REMINDERS FOR ELECTRICAL SAFETY OFFICE OF HEALTH & SAFETY		
1	**Check electrical appliances, cables, and power sockets regularly for defects.** Don't use devices with frayed cords, broken plugs, or damaged insulation.	 
2	**Do not overload wall sockets or extension cords.** Avoid using multi-way adaptors with extension sockets. Never daisy-chain extension cords together, as the wall socket can become overloaded.	
3	**Switch off the wall socket and unplug appliances when not in use.** This saves energy and reduces the risk of shock and fire.	
4	**Keep electrical appliances away from water.** Never use electrical devices or touch an electrical outlet with wet hands or in damp conditions.	
5	**Use only equipment and appliances with the Singapore SAFETY Mark.** The SAFETY Mark shows that the product meets Singapore's safety regulation standards.	SAFETY MARK 123456-00

Safety - Tools

If you’re new with the use of these tools, please get assistance from the laboratory technician to ensure safety.



Safety Precautions – Wire Stripper

#1 Disconnect the wire from the power source.

- Never Strip a LIVE Wire.

#2 Do not use a wire stripper with damaged, rusty, worn-out grips or dented blades.

3 Keep fingers away from the sharp blades.

#4 Use the correct slot to prevent damaging the wire or leaving insulation uncut.

#5 Hold the wire firmly to prevent slipping while stripping.

Watch Video for proper use of wire stripper

https://youtu.be/Mma4GaB63ro

Safety Precautions – Long nose plier

#1 Inspect Before Use

- Check for cracks, rust, or worn-out grips.
- Ensure jaws are aligned and open/close smoothly.

#2 Use the Right Tool

- Only use long nose pliers for gripping, bending, or cutting small wires.
- **Do not use as a hammer, pry bar, or wrench.**

3 Grip Properly

- Hold the pliers firmly by the handle.
- Keep fingers away from the cutting edge and moving parts.

#4 Avoid Excessive Force

- Do not apply too much force—this may cause the pliers to slip or snap.

Watch video for proper use of long nose plier

https://youtu.be/srsPwAWC2MQ

Safety Precautions – IC-Extractor or Puller

#1 Power Off the Device

- Always switch off and unplug the device before removing ICs to prevent electrical shock or *circuit damage*.

***# 2 Inspect the Extractor* Tool**

- Check the extractor tool for bent tips, rust, or signs of wear. A damaged tool may slip and cause injury or damage the IC.

#3 Ensure Proper Lighting

- Work in a well-lit area to clearly see the IC orientation and alignment, reducing the risk of error or component damage.

Watch video for proper use of IC-Puller

https://youtu.be/IHyoTo06jwo

Safety – Drill Press Machine

Users are required to:

- Watch the safety video, and
- Achieve a minimum pass score of 80% in the quiz.

Users are permitted to operate the Drill Press Machine only after meeting these requirements.

Click link to watch a short video of Drill Press Machine

https://youtu.be/lpXtq_wMxoQ

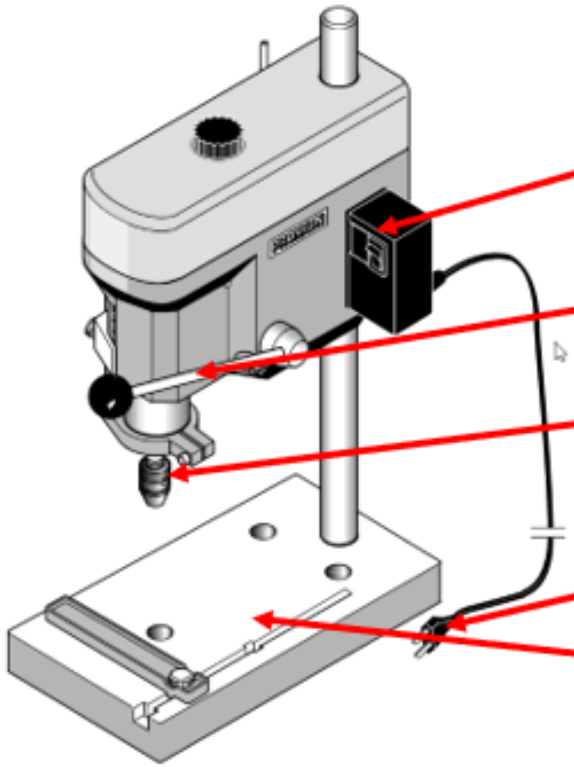
Safety Precautions –Drill Press Machine

- **Wear** safety goggles/PPE
- **Tie back** long hair and avoid loose clothing (roll up sleeves).
- **Ensure** the work area is well-lit and free of clutter.
- **Ensure** that the drill bit is locked securely in the chuck.
- **STOP IMMEDIATELY** if the drill press is malfunctioning and report the issue to the Technician or Supervisor.
- **ALWAYS** keep the working surface clean and organized.

Safety Precautions – Drill Press Machine

- **NEVER** set up or adjust the drilling machine while it's running
- **NEVER** wear loose clothing or jewellery.
- **NEVER** hold the workpiece to drill them; secure them to the with a vise or clamp.
- **NEVER** use tools that are in poor condition
- **NEVER** try to stop the drill chuck and spindle with your hands
- **NEVER** clean the machine while it's running.

Know the Control



ON/OFF Switch

Handle screw - lift or lower the drill bit

Chuck to hold the

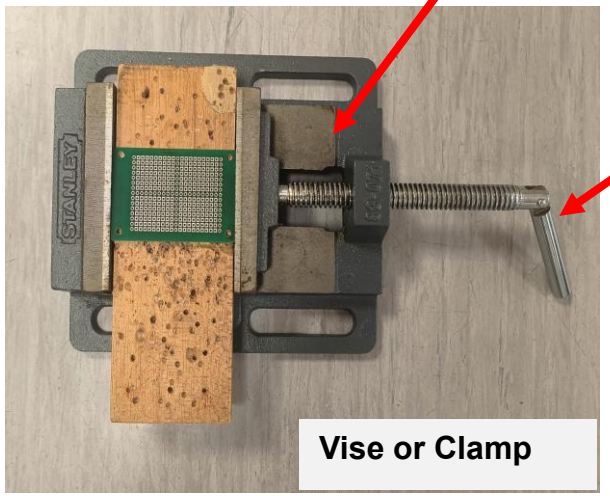
Main power

Worktable – placed workpiece secure with the vise or clamp

Safety Goggles



Workpiece



Vise or Clamp

Turn lever clock-wise direction to tighten workpiece

Safety - Soldering Iron

Users are required

- Watch the safety video, and
- Achieve a minimum pass core of 80% in the quiz.

Users are permitted to operate the Soldering Tool only after

meeting these requirements.

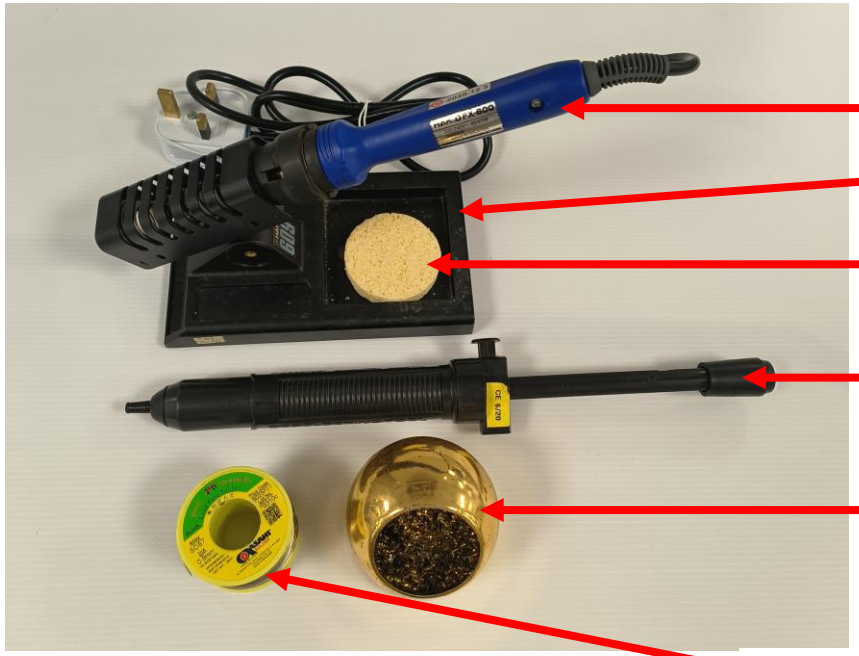
Click link to watch video – operate the soldering iron

https://youtu.be/Fi6cSJ8ijeo

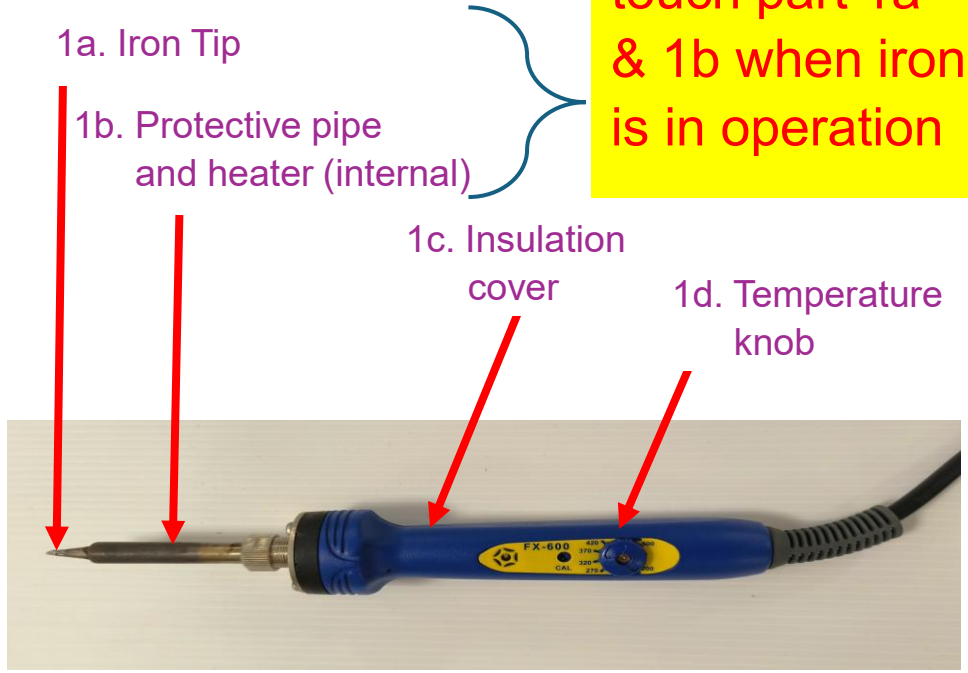
Safety Precautions – Soldering Iron

- **NEVER** touch the tip of the soldering iron. It is very hot and will burn.
- **NEVER** use soldering iron that are in poor condition.
- **Hold wires to be heated** with tweezers, pliers or clamps to avoid receiving burns from objects that are heated.
- **Always placed the iron** into the stand when not in use.
- **Use lead free solder**.
- Keep the **cleaning sponge wet** during use.
- **Must turn on** the fume extractor during soldering activities.

Know the Soldering Parts and Accessories

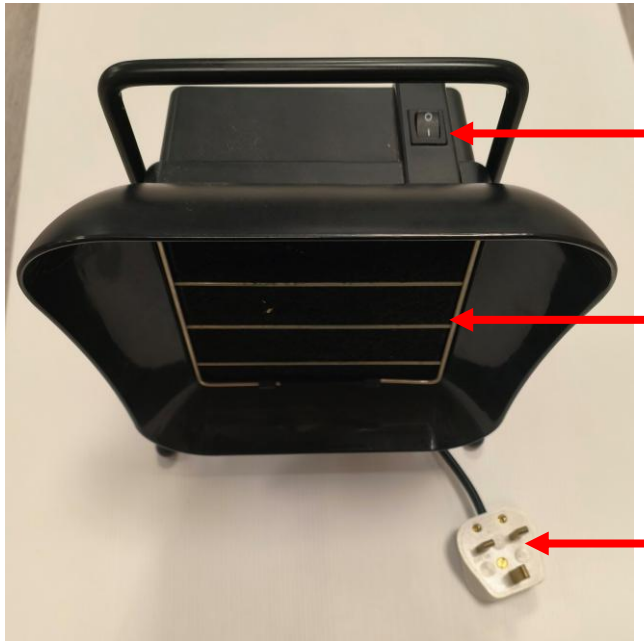


HOT, Do not touch part 1a & 1b when iron is in operation



1.Soldering iron – heat and melt solder to join two or more wires
2. Iron stand – hold the solder iron
3. Sponge – remove or clean excess solder from the iron tip
4. Sucker Tool – Use to remove unwanted solder when desoldering/disassemble parts
5. Brass wool or mesh – same function as sponge
6. Lead free solder –better for health and environment
7. Power source

Know the Soldering Parts and Accessories

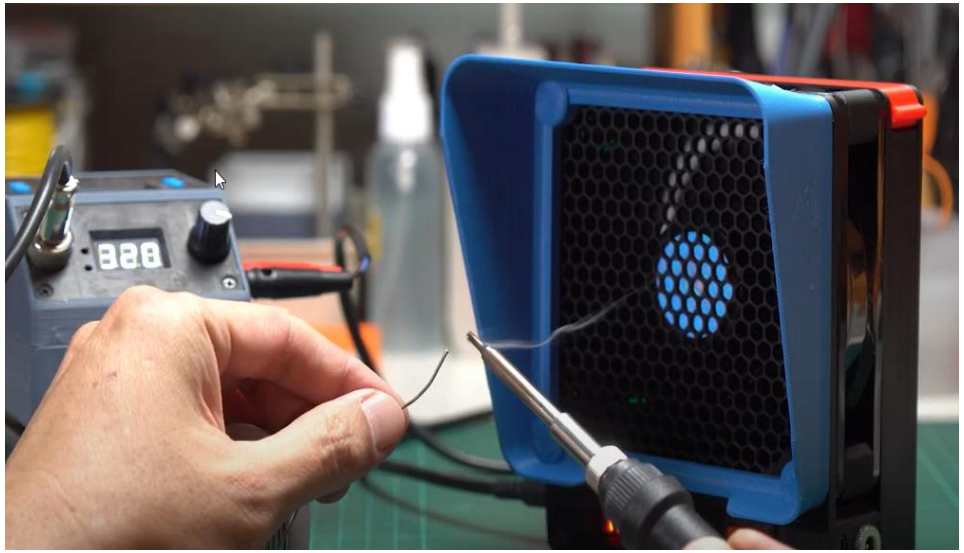


1. “ON/OFF” switch

2. Fume absorption surface

3. Power plug

The picture below shows how to position the fume extractor.



1. On/Off switch – To power ON and OFF the Fume extractor

2. Fume absorption surface – pull in fumes and dust particular into a contained filtration system.

3. Power plug – plug into electrical socket to supply the electrical power to the fume extractor.